Taha Ameen

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Research Interests

Matching Markets, Applied Probability, Statistical Inference, Dynamic Matching, Stochastic Networks, Combinatorial Optimization

Academic Appointments

Carnegie Mellon University *Sep* 2026 – *Present*
Postdoctoral fellow
School of Computer Science
Advisor: Prof. Weina Wang

Education

University of Illinois Urbana-Champaign *Sep* 2020 – *Aug* 2026
PhD in Electrical and Computer Engineering
Thesis: "Fundamental Limits of Network Inference and Matching"
Advisor: Prof. Bruce Hajek

American University of Sharjah *Sep* 2015 – *Dec* 2019
BS in Electrical Engineering
BS in Mathematics

Internships

Nokia Bell Labs *Jun* 2022 – *Aug* 2022
Research Intern
Murray Hill, NJ, USA

Aalto University *Jun* 2019 – *Aug* 2019
Research Intern
Espoo, Finland

Awards and Fellowships

Paper Awards and Finalist Distinctions

- INFORMS MSOM Best Student Paper Prize, Finalist (Winner TBA) 2026
- Stochastic Networks Conference Outstanding Student Poster Award 2026
- Jack Keil Wolf ISIT Student Paper Award 2025
- IISA Best Student Paper Award in Applications of Statistics and Data Science, Finalist 2025
- INFORMS APS Best Student Paper Prize, Finalist 2024

Fellowships

- A.R. "Buck" Knight Fellowship 2025
- MAVIS Future Faculty Fellowship 2024
- Robert T. Chien Memorial Award in Electrical Engineering 2024
- Joan and Lalit Bahl Fellowship 2023, 2022
- James M. Henderson Fellowship 2021

Travel Grants

- INFORMS Applied Probability Society Travel Grant 2025
- IEEE International Symposium of Information Theory Travel Grant 2025
- International Indian Statistics Association Travel Grant 2025

Selected Talks and Posters

Upcoming

A uniformity principle for spatial matching.
INFORMS Annual Meeting, Session on Auctions and Market Design 2026
INFORMS Annual Meeting, MSOM Award Session 2026

Recent progress on flexibility design in random matching markets.
IE Graduate Seminar, University of Pittsburgh 2026

Past

Recent progress on alignment and detection of multiple correlated random graphs.
CMU Theory Lunch Seminar [Video] 2026

A uniformity principle for spatial matching.
INFORMS Revenue Management and Pricing Section Conference 2026
INFORMS MSOM Conference, Session on Network Design and Spatial Matching 2026
Stochastic Networks Conference 2026
Marketplace Innovation Workshop (MIW) 2026

Flexibility allocation in random bipartite matching markets: exact matching rates and dominance regimes.
Rotman Young Scholar Seminar Series, University of Toronto 2026

Aligning Multiple Inhomogeneous Random Graphs: Fundamental Limits of Exact Recovery.
INFORMS Annual Meeting, Session on Applied Probability in Algorithmic and Operational Networks II 2025
IISA Best Student Paper Award Talk, Statistics and Data Science Category 2025

Impacts of network heterogeneity on random graph matching.
INFORMS Applied Probability Society Conference, Session on High Dimensional Statistics 2025

Exact random graph matching with multiple graphs.
Cornell ORIE Young Researchers Workshop 2025
IEEE International Symposium on Information Theory 2025
INFORMS APS Best Student Paper Award Competition 2024
INFORMS Annual Meeting, APS Session on Theoretical Advances in Networks, Dynamics and Inference 2024
IDEAL Workshop at Northwestern University 2024
Stochastic Networks Conference 2024

Robust graph matching when nodes are corrupt.
International Conference on Machine Learning 2024
Coordinated Science Lab Student Conference at UIUC 2024

Blockchain security when messages are lost.
ACM Workshop on Developments in Consensus 2022

Earlier presentations

IEEE Wireless Telecommunications Symposium 2020
Allerton Conference 2019

Publications and Preprints

Preprints

9. T. Ameen and W. Wang, "Achieving Optimal Scaling of Queue Size for an Input-Queued Switch", *Submitted*.
8. T. Ameen and S. H. Yu, "Distance flexibility in spatial matching: the value of concentration". [arXiv]
7. T. Ameen and B. Hajek, "Converse bounds for multiple graph alignment and correlation detection based on last matching", *Submitted*. [arXiv]
6. T. Ameen and N. Maitra, "Estimating Community Boundaries in Geometric Random Graphs", *Submitted*. [arXiv]
5. T. Ameen and B. Hajek, "Sharp Detection Threshold for Correlation among Multiple Unlabeled Gaussian Networks", *Submitted*. [arXiv]
4. W. Wang, T. Ameen, Y. Chen, Y. Hong, J. Nichols and M. Zurek, "Optimality of a Threshold Policy for a Queueing System with One Fast Server and Two Identical Slow Servers", *Submitted*. [arXiv] [1]
3. T. Ameen, F. Sentenac and S. H. Yu, "Flexibility allocation in random bipartite matching markets: exact matching rates and dominance regimes", *Submitted*. [arXiv]
2. T. Ameen, F. Sentenac and S. H. Yu, "A Uniformity Principle for Spatial Matching", *Submitted*. [arXiv]
   - Winner of outstanding student poster award at Stochastic Networks Conference, 2026.
   - Finalist for INFORMS MSOM Best Student Paper Competition, 2026.
1. T. Ameen, K Kytölä and S.C. Park, "Slit-strip Ising boundary conformal field theory 2: Scaling limits of fusion coefficients", *Submitted*. [arXiv]

Journal Publications

7. T. Ameen and B. Hajek, "Aligning Multiple Inhomogeneous Random Graphs: Fundamental Limits of Exact Recovery", *Operations Research*, 2026. [arXiv]
   - Finalist for best student paper award in applications of statistics and data science at IISA 2025.
6. T. Ameen and B. Hajek, "Detecting Planted Structure in Circular Data", *IEEE Transactions on Information Theory*, 2026. [arXiv]
5. T. Ameen, S. Mukhopadhyay and N. Qaddoumi, "Computing robust forward invariant sets of multidimensional nonlinear systems via geometric deformation of polytopes", *IEEE Transactions on Automatic Control*, 2023. [Link]
4. T. Ameen, K. Kytölä, S.C. Park and D. Radnell, "Slit-strip Ising boundary conformal field theory 1: Discrete and continuous function spaces", *Mathematical Physics, Analysis and Geometry*, Springer, 2022. [Link]
3. S. Shahriar, J. Ramesh, A. Towheed, T. Ameen, A. Sagahyroon and A. Al-Ali, "NICE: Narrative Integrated Career-Exploration Platform", *Frontiers in Education*, 2022. [Link]
2. T. Ameen, M. Hasan and M. Ismail, "A Novel Medium Access Control Algorithm for Ad Hoc Networks based on Ising Model", *IEEE Access*, 2020. [Link]
1. T. Ameen, Y. Aborahama, M. Hasan and M. Ismail, "A PDE-based approach for the evaluation of probability of starvation in video streaming", *Physical Communication*, Elsevier, 2020. [Link]

Conference Publications

8. T. Ameen and B. Hajek, "Detecting Planted Structure in Circular Data", *IEEE International Symposium on Information Theory (ISIT),* 2026. [Link]
7. T. Ameen and B. Hajek, "Detecting correlation between multiple unlabeled Gaussian networks", *IEEE International Symposium on Information Theory (ISIT),* 2025. [Link]

[1] The core technical ideas in this paper were generated by GPT-5.5 Pro; the authors verified the proofs and rewrote the paper to improve rigor, clarity, and exposition.

6. T. Ameen and B. Hajek, "Exact random graph matching with multiple graphs", *IEEE International Symposium on Information Theory (ISIT),* 2025. [Link]
   - Winner of Jack Keil Wolf ISIT Student Paper Award, 2025.
   - Finalist for INFORMS APS Best Student Paper Competition, 2024.
5. T. Ameen and B. Hajek, "Robust graph matching when nodes are corrupt", *International Conference on Machine Learning*, 2024. [Link]
4. T. Ameen, S. Sankagiri and B. Hajek, "Blockchain security when messages are lost", *ACM Workshop on Developments on Consensus*, 2022. [Link]
3. UIUC Info Theory Students, S. Basu and L. Varshney, "The twelvefold way of non-sequential lossless compression", *IEEE Data Compression Conference*, 2021. [Link]
2. T. Ameen, M. Hasan and M. Ismail, "A queue-length based approach to metropolized Hamiltonians for distributed scheduling in wireless networks", *IEEE Wireless Telecommunications Symposium*, 2020. [Link]
1. T. Ameen, S. Mukhopadhyay and S. Farhana, "A Novel Expression for Computing Time Response of LTI Systems of Arbitrary Order with Applications to Fractional and Stochastic Control", *Allerton Conference on Communication, Control and Computing*, 2019. [Link]

Teaching and Mentoring

- Guest Lecturer: CS 15-359: Probability and Computing II, Carnegie Mellon University, *Fall '*26.
- Guest Lecturer: ORI 397: High Dimensional Statistics and Probability, UT Austin, *Fall '*26.
- Teaching Assistant: ECE 543 Statistical Learning Theory, UIUC, *Spring '*25.
- Teaching Assistant: ECE 534 Random Processes, UIUC, *Spring '*24*, Fall '*24.
- Undergraduate tutor, IEEE Eta Kappa Nu Honor Society, 2018-19.
- Peer mentor for undergraduate students in engineering, Academic Support Center, 2016-19.

Service

Reviewing.

- Journals: Mathematics of Operations Research, Annals of Applied Probability, IEEE Transactions on Systems, Man and Cybernetics: Systems, Wiley International Journal of Robust and Nonlinear Control, IEEE Sensors.
- Conferences: NeurIPS, ACM Sigmetrics, IEEE International Symposium on Information Theory (ISIT), IEEE Conference on Decision and Control (CDC), IEEE Information Theory Workshop (ITW).

Volunteering.

- INFORMS MSOM Conference, 2026.
- IEEE International Symposium on Information Theory, 2025.
- CSL Student Conference, 2022.
- Allerton Conference, 2019, 2022, 2023.